Project 3: AC Load Measurement

Malachy Crossan & Jacob Rosen

EEL3123-C0013: Linear Circuits II

Due Date:	2026-07-14
Last modified:	2026-07-14

Contents

3.A Objective

Box #1

The objective of Project 3 was to determine the impedance and power factor of a concealed AC load without opening the box or directly measuring its internal resistance, capacitance, or inductance. The unknown load could be a purely resistive, RL, or RC network.

Equipment:

- Hantek 3-in-1 digital equipment
- Digital multimeter (DMM)
- Powered breadboard and leads
- Reference resistor
- Concealed AC load box (Unknown RR, RL, or RC network)
- Oscilloscope
- USB storage (USB screenshots required by the lab guidelines)

3.B Plan

A known reference resistor was connected in series with the concealed box so that the circuit current could be calculated from a voltage measurement.

- Construct a safe series test circuit containing the source, a selected reference resistor, and the concealed load.
- Measure the RMS voltages across the source, reference resistor, and concealed load.
- Use the reference-resistor voltage to calculate circuit current. ($\mathrm{Irms} = \frac{\mathrm{Vrms}}{R}$)
- Measure the phase difference between the load voltage and reference-resistor voltage with the oscilloscope.
- Calculate load impedance, rectangular components, and power factor. then classify the load as RR, RL, or RC.

3.C Results

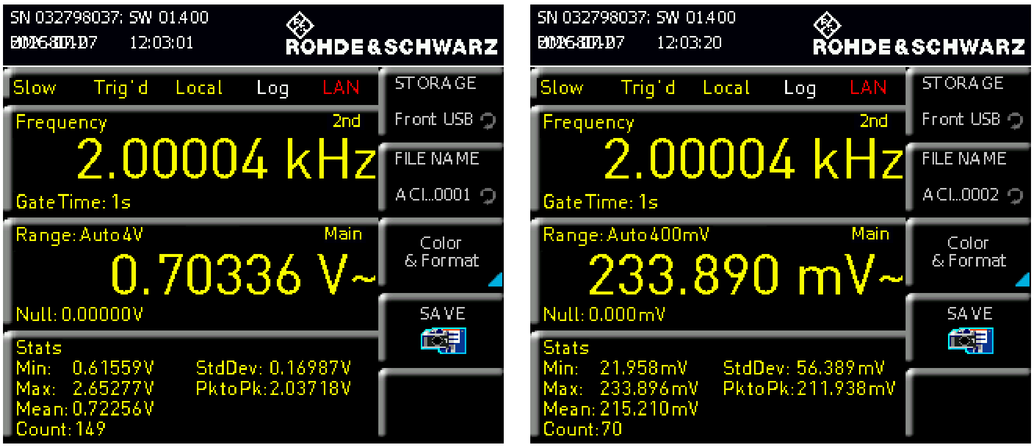

Figure 1: DMM measurements. V_s (left) and V_2 (right) RMS voltages.

With V_s peak to peak set to $2V$ at $2k\mathrm{Hz}$, we measured the following RMS voltages with our DMM:

$$\begin{aligned} V_s &= 703mV \\ V_2 &= 234mV \end{aligned} \qquad \text{Eq 1}$$

The $10k\Omega$ (measured $9.92k\Omega$) resistor was used to determine the current through the unknown load box. Using the DMM measurements, the resistor RMS voltage was $234mV$, giving a current of:

$$\text{Irms} = \frac{0.234}{992} = 0.0236mA \qquad \text{Eq 2}$$

The voltage across the load box was approximately:

$$V_1 = 0.703 - 0.234 = 0.469V \qquad \text{Eq 3}$$

3.C.1 Load Impedance

Therefore, the impedance was approximately:

$$Z_1 = \frac{0.469V}{0.0236mA} = 19.8k\Omega \qquad \text{Eq 4}$$

The oscilloscope measurements in Figure 2 show $467.79mV$ RMS across the box and $230.52mV$ RMS across the resistor. These measurements produced an impedance of approximately $20.1k\Omega$, which closely agreed with the DMM calculation.

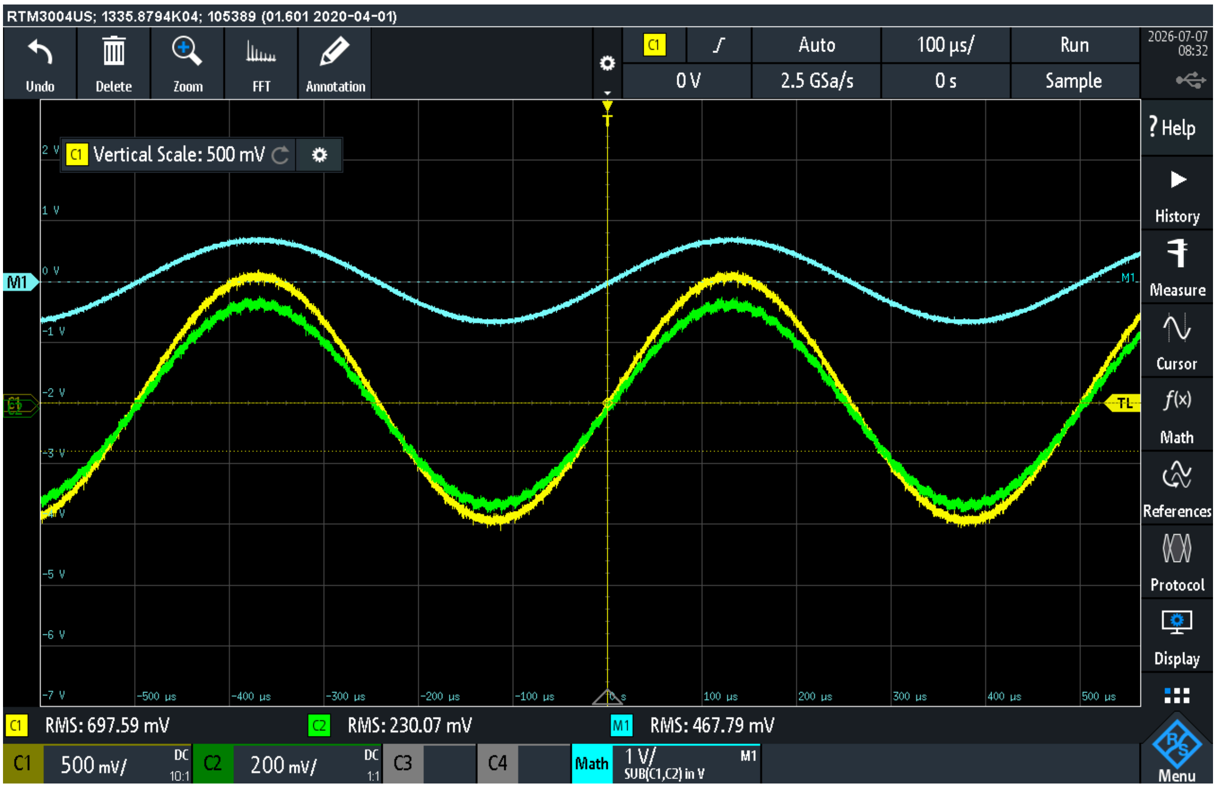

Figure 2:

C1 probe: V_s (Positive terminal of box to Reference ground)
C2 probe: V_2 (Negative terminal of box to Reference ground)
M1 calc: V_1 (Positive terminal to negative terminal of box)

3.C.1.a Rectangular Components

The rectangular components of the load impedance were calculated as follows:

$$
\begin{gathered}
Z_1 = 19.8k\Omega \\
\mathrm{Irms}_1 = 23.6\mu A \\
\mathrm{Vrms}_1 = 0.469V \\
\mathrm{Real} = Z_1 \cos(1.19°) = 19.8k\Omega \\
\mathrm{Imaginary} = Z_1 \sin(1.19°) = 0.412k\Omega
\end{gathered}
\qquad \text{Eq 5}
$$

3.C.2 Phase Angle and Power Factor

3.C.2.a Analytical Approach

We noticed that the load voltage was very slightly lagging so we decreased our source frequency to $100\ \mathrm{Hz}$ to make the phase difference more noticeable.

The Δt was then measured with the oscilloscope by comparing the time the reference resistor voltage and the load box voltage crossed $0V$. The oscilloscope measurements in Figure 3 show a Δt of $3.90ms$.

Figure 3:

C2 probe: V_2 (Load box voltage)
M1 calc: V_1 (Positive terminal to negative terminal of box)

$$
\begin{gathered}
\Delta t = 3.90ms \\
\mathrm{f} = 100\ \mathrm{Hz} \\
\mathrm{T} = \frac{1}{f} = 10ms \\
\text{Phase Angle @}\mathbf{100\ Hz} = \frac{\Delta t}{T} \times 360° = \frac{3.90ms}{10ms} \times 360° = 140.4°
\end{gathered}
\qquad \text{Eq 6}
$$

Importantly, this phase angle was measured at $100\ \mathrm{Hz}$, but the load box was designed to operate at $2k\ \mathrm{Hz}$ so this only helped us determine that the load is in fact inductive.

3.C.2.b Measured Approach

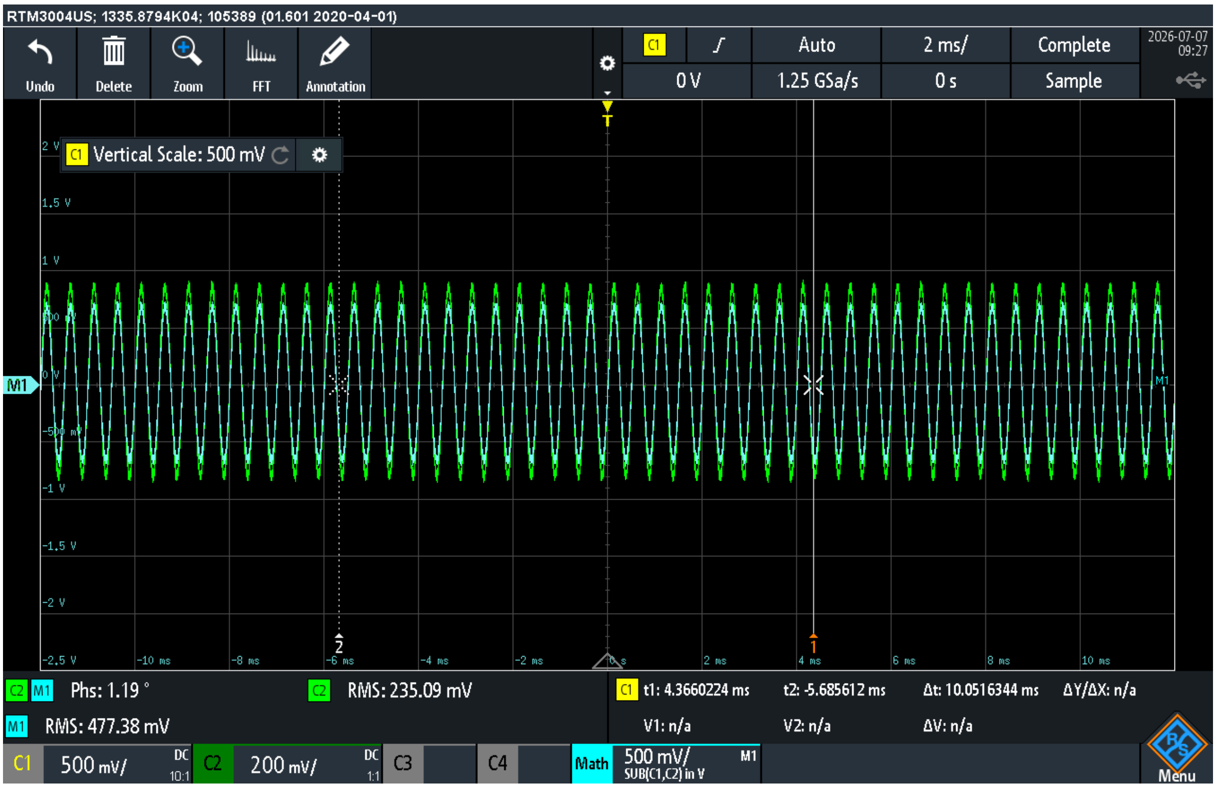

Figure 4:

C2 probe: V_2 (Load box voltage)
M1 calc: V_1 (Positive terminal to negative terminal of box)

The measured phase angle (Figure 4) was only $1.19°$ giving a power factor of:

$$\mathrm{PF} = \cos(1.19°) = 0.9998 \qquad \text{Eq 7}$$

3.D Conclusion

The objective of this experiment was to determine the impedance and type of an unknown AC load by measuring RMS voltages and the phase difference between voltage and current. The load-box impedance was found to be approximately $20k\Omega$, and the phase angle was measured as about $1.19°$.

Since the phase difference was lagging, the unknown box was classified as an RL load.

This experiment demonstrated how a known series resistor can be used to calculate AC current and unknown impedance. It also showed how oscilloscope phase measurements can distinguish between resistive, inductive, and capacitive loads. These methods will be useful in future AC circuit analysis and troubleshooting.